

Advanced Design Project

A Comprehensive Review of Runway Incursion and Accident Conflict Geometries



TECHNISCHE
UNIVERSITÄT
DARMSTADT

Students: Mikaela Aleinikova (2554541)
Hardik (2791472)
Jonathan David Magsino (2358268)
Rodrigo David Loza Egoavil (2367521)
Sajad Jafari Kichi (2275545)

Department: Mechanical Engineering / FB 16

Focus: Aerospace Engineering

Supervisors: Pascal Menner, M.Sc., Stefan Neis, M.Sc. (external)

Background:

Runway incursions include occurrences of an unauthorized aircraft operating in the protected area of an active runway (e.g., one on which another aircraft attempts to land), potentially creating a collision hazard. In recent years, multiple near-miss incidents were reported due to runway incursions. Boeing is therefore developing new capabilities to reduce runway incursions and avoid collision, including a flight deck-based system that alerts pilots against other aircraft operating on their active runway.

Task:

The goal of this ADP is to identify, characterize, and classify runway incursions (and runway accidents) of the last 30 years to extract typical patterns against which Boeing's new system must alert. A comprehensive analysis is to be conducted to identify the most frequently occurring runway incursions and their causes. These are to be characterized by the runway layout (i.e., independent runway vs. intersecting runways), conflict geometry (i.e., which operations the conflicting aircraft were performing, such as takeoff, taxi, or landing), and cause (e.g., controller error, pilot error, equipment failure, etc.). Next, the industry standard (RTCA DO-323) alert behaviors are to be applied to the identified datasets to determine their effectiveness in preventing collisions. Finally, recommendations to increase alert effectiveness in avoiding collisions are to be derived and evaluated.

Work Packages:

The scope of this thesis shall include:

- Literature research on runway incursions and surface collision alerting
- Identification of all relevant runway incursions and accidents since 1995 (categories A and B) globally, primarily at commercial and cargo service airports (though not necessarily limited to those)
- Characterization and classification of runway incursions (e.g., runway layout, geometry, and cause) and presentation of overall statistics
- Application of RTCA DO-323 alerting behaviors onto the identified datasets to evaluate the effectiveness of that alerting design
- Derive and evaluate recommendations to improve collision avoidance alert effectiveness based on parameter adjustments
- Documentation (English) and presentation of results.

Darmstadt, den 16.10.2024

Prof. Dr.-Ing. U. Klingauf

Institut für Flugsysteme und
Regelungstechnik

Institute of Flight Systems and
Automatic Control



Prof. Dr.-Ing. Uwe Klingauf

Otto-Berndt-Str. 2
64287 Darmstadt

Sekretariat:
Katja Möller
+49 6151 16 - 21044
moeller@fsr.tu-darmstadt.de

Betreuer:
Pascal Menner, M.Sc.
+49 6151 16 - 21072
menner@fsr.tu-darmstadt.de